Left-Recursion

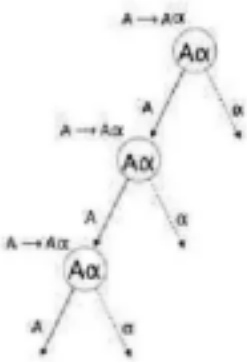
1. Left-Recursion $S \rightarrow Sa / \in$
2. Right-Recursion $S \rightarrow aS / \in$
3. General-Recursion $S \rightarrow aSb / \in$

A production of grammar is said to have left recursion if the leftmost variable of its RHS is same as variable of its LHS.

A grammar containing a production having left recursion is called as Left Recursive Grammar.

Left-Recursion

$A \rightarrow A\alpha / \beta$



Left recursion is considered to be a problematic situation for Top down parsers.

Therefore, left recursion has to be eliminated from the grammar.

Left-Recursion Elimination

we can eliminate left recursion by replacing the pair of productions with-

$A \rightarrow \beta A'$

$A' \rightarrow \alpha A' / \in$

Problem-01:

Consider the following grammar and eliminate left recursion-

$A \rightarrow ABd / Aa / a$

$B \rightarrow Be / b$

Solution-

The grammar after eliminating left recursion is-

$A \rightarrow aA'$

$A' \rightarrow BdA' / aA' / \in$

$B \rightarrow bB'$

$B' \rightarrow eB' / \in$

Problem-02:

Consider the following grammar and eliminate left recursion-

$E \rightarrow E + E / E \times E / a$

Solution-

The grammar after eliminating left recursion is-

$E \rightarrow aA$

$A \rightarrow +EA / xEA / \in$

Problem-03:

Consider the following grammar and eliminate left recursion-

$E \rightarrow E + T / T$

$T \rightarrow T \times F / F$

$F \rightarrow id$

Solution-

The grammar after eliminating left recursion is-

$E \rightarrow TE'$

$E' \rightarrow +TE' / \in$

$T \rightarrow FT'$

$T' \rightarrow xFT' / \in$

$F \rightarrow id$

Problem-04:

$S \rightarrow (L) / a$

$L \rightarrow L , S / S$

Solution-

The grammar after eliminating left recursion is-

$S \rightarrow (L) / a$

$L \rightarrow SL'$

$L' \rightarrow ,SL' / \in$

Consider the following grammar and eliminate left recursion-

S → S0S1S / 01

Solution-

The grammar after eliminating left recursion is-

S → 01A

A → 0S1SA / ∈

Consider the following grammar and eliminate left recursion-

S → A

A → Ad / Ae / aB / ac

B → bBc / f

Solution-

S → A

A → aBA' / acA'

A' → dA' / eA' / ∈

B → bBc / f

A → Ba / Aa / c

B → Bb / Ab / d

Step-01:

First let us eliminate left recursion from A → Ba / Aa / c

A → BaA' / cA'

A' → aA' / ∈

Now, given grammar becomes-

A → BaA' / cA'

A' → aA' / ∈

B → Bb / Ab / d

Step-02:

Substituting the productions of A in B → Ab, we get the following grammar-

A → BaA' / cA'

A' → aA' / ∈

B → Bb / BaA'b / cA'b / d

Step-03:

Now, eliminating left recursion from the productions of B, we get the following grammar-

$A \rightarrow BaA' / cA'$

$A' \rightarrow aA' / \in$

$B \rightarrow cA'bB' / dB'$

$B' \rightarrow bB' / aA'bB' / \in$

FIRST and FOLLOW

First and Follow sets are needed so that the parser can properly apply the needed production rule at the correct position.

FIRST

First(α) is a set of terminal symbols that begin in strings derived from α.

Consider the production rule-

$$A \rightarrow abc / def / ghi$$

Then, we have- First(A) = { a , d , g }

FIRST and FOLLOW

FIRST

For a production rule $X \rightarrow \in$,

First(X) = { ∈ }

For any terminal symbol 'a',

First(a) = { a }

FIRST and FOLLOW

FOLLOW

Follow(α) is a set of terminal symbols that appear immediately to the right of α.

- For the start symbol S, place $ in Follow(S).
- For any production rule A → αB,

  Follow(B) = Follow(A)
- For any production rule $A \rightarrow \alpha B \beta$,

  If $\in \notin$ First(β), then Follow(B) = First(β)

  If $\in \in$ First(β), then Follow(B) = { First(β) – $\in$ } ∪ Follow(A)

FIRST and FOLLOW

$\in$ may appear in the first function of a non-terminal.

$\in$ will never appear in the follow function of a non-terminal.

Before calculating the first and follow functions, eliminate Left Recursion from the grammar, if present.

PRACTICE PROBLEMS BASED ON CALCULATING FIRST AND FOLLOW-

Problem-01:

Calculate the first and follow functions for the given grammar-

$$S \rightarrow aBDh$$
$$B \rightarrow cC$$
$$C \rightarrow bC / \in$$
$$D \rightarrow EF$$
$$E \rightarrow g / \in$$
$$F \rightarrow f / \in$$

Solution-

The first and follow functions are as follows-

First Functions-

- First(S) = { a }
- First(B) = { c }
- First(C) = { b , $\in$ }
- First(D) = { First(E) – $\in$ } ∪ First(F) = { g , f , $\in$ }
- First(E) = { g , $\in$ }
- First(F) = { f , $\in$ }

Follow Functions-

- Follow(S) = { $ }
- Follow(B) = { First(D) – $\in$ } ∪ First(h) = { g , f , h }
- Follow(C) = Follow(B) = { g , f , h }
- Follow(D) = First(h) = { h }
- Follow(E) = { First(F) – $\in$ } ∪ Follow(D) = { f , h }

LL (Parsing table